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Machine learning-based optimization of enhanced nitrogen removal in a full-scale urban wastewater treatment plant with ecological combination ponds

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ABSTRACT

Urban wastewater treatment plants (WWTPs) employing ecological combination ponds (ECPs) in small towns across China are struggling to dynamically adjust operating parameters under fluctuating influent quality conditions, resulting in excessive aeration and external carbon source overdosing to meet the stringent total nitrogen (TN) discharge standards. To address these issues, it is essential to establish an appropriate enhanced TN removal model for WWTPs with ECPs. In this study, we collected three years of operational data from a full-scale urban WWTP utilizing ECPs and implemented an interpretable machine learning approach to predict and optimize effluent TN concentration. The XGBoost model attained R^2 values of 0.997 and 0.911, RMSE values of 0.196 and 1.283 for the training and testing sets, respectively. The Shapley additive explanation analysis and partial dependence plots identified optimal operating parameters to improve TN removal while balancing energy consumption and chemical oxygen demand (COD) dosage reduction. A graphical user interface was developed to facilitate ongoing prediction and coordinated optimization of process operational parameters, achieving simultaneous reductions in effluent TN, energy consumption, and external carbon source usage. Notably, the effluent TN concentration decreased by 17.50 %, while COD dosage was reduced by 33.29 % annually. Consequently, WWTP with ECPs demonstrated substantial potential for carbon emission reduction. Total annual carbon emission reductions (788.40 t CO_2/y) were calculated solely based on enhanced TN removal, along with reductions in energy consumption and COD dosage. Our findings provide the optimal model for enhanced TN removal in urban WWTPs utilizing ECPs under variable influent quality pressure, thereby meeting stringent TN discharge standards and contributing to energy savings and carbon emission reduction goals.

1. Introduction

Ecological combination ponds are sewage treatment systems that integrate biological and ecological treatment processes, characterized by minimal surplus sludge production and no sludge recirculation. These ponds feature vertical and multi-level mode with ecological plant floating beds on the sewage surface, effectively combining sewage treatment with landscape construction (Pan et al., 2023). In small towns across China, ecological combination ponds have been employed for treating domestic sewage due to their low cost and ecological benefit (Huang et al. 2016). However, urban wastewater treatment plants (WWTPs) utilizing ecological combination ponds are currently struggling to dynamically adjust operating parameters due to the irregular discharge of industrial wastewater, resulting in excessive aeration and external carbon source overdosing under fluctuating influent quality conditions to meet the stringent total nitrogen discharge standards (Rojas et al. 2020; Bhatt et al. 2022). To address these issues, it is necessary to establish an appropriate enhanced total nitrogen removal model for urban WWTPs with ecological combination ponds (Gallego-Schmid et al. 2019; Chen et al. 2018).

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The rapid advancements in data science and artificial intelligence have facilitated the gradual adoption of data-driven machine learning (ML) methods for predicting and optimizing wastewater treatment processes (Rios Fuck et al. 2024; Ye et al. 2024; Lv et al. 2024; Zaghloul et al. 2022; Rizal et al. 2022; Jiang et al. 2024). As a result, a series of ML models, including long short-term memory (LSTM), support vector regression (SVR), random forest (RF), artificial neural network (ANN), and extreme gradient boosting (XGBoost), have been developed to predict and improve effluent quality (Heo et al. 2021; Pan et al. 2024; Emaminejad et al. 2023; Wang et al. 2024; Wang et al. 2023). For instance, Pan et al. established a ML-assisted high-throughput method to rapidly identify and optimize mixed carbon sources. They employed the XGBoost algorithm to predict the total nitrogen removal rate and microbial growth, thereby aiding in the assessment of the denitrification potential. However, their study was conducted using synthetic wastewater, which limited effective guidance for practical scenarios in wastewater treatment plants (Pan et al. 2024). Wang et al. developed the variation sliding layer (VSL)-ML models to predict and control air demand precisely of wastewater treatment plants, showing a 16.12 % reduction in air demand compared to conventional aeration control of preset dissolved oxygen and feedback to the blower. Although the VSL-ML models demonstrated great potential to be applied for precision air demand prediction and control, they focused on the performance of a single indicator (Wang et al. 2023). Overall, there remains a research gap in establishing machine learning models that enhanced nitrogen removal of full-scale wastewater treatment plants from optimization of key process parameters.

A previous study indicated that optimizing nitrogen removal was sensitive to variations in influent quality conditions and conflicted with energy-saving objectives (Fernández-Arévalo et al. 2017). Consequently, ML approaches have leveraged historical data to build accurate predictive models and optimize operational parameters, focusing on handling complex process variations and variable influent conditions in full-scale WWTPs (Zhong et al. 2021; Hao et al. 2015; Qambar et al. 2022; Song et al. 2024; Shao et al. 2013; Luo et al. 2023; Chen et al. 2023; Li et al. 2022; Chen et al., 2024a). For example, Qambar et al. proposed a dynamic supervised ML model that relied on real-time influent biochemical oxygen demand prediction and biological treatment oxygen limitation, which resulted in a 23 % reduction in energy consumption of a municipal wastewater treatment plant (Qambar et al. 2022). Recently, a process time-lag simulation-enhanced machine learning model was developed, taking a full-scale wastewater treatment plant with Anoxic/Anaerobic/Aerobic process as a case study, achieving R^2 values of 0.94 and 0.79 for the training and testing sets, respectively. It enabled the simulation and causal interpretation of nitrogen removal, while optimizing nitrogen removal under multi-objective scenarios to achieve low-carbon and stable operation of wastewater treatment plant (Chen et al., 2024b). Nevertheless, appropriate machine learning models have not been established to predict and optimize nitrogen removal in full-scale urban WWTPs employing ecological combination ponds under fluctuating influent quality conditions. In particular, achieving enhanced nitrogen removal while balancing energy consumption and external carbon source usage reduction remains limited.

To address this, we collected three years of operational data from a full-scale urban wastewater treatment plant utilizing ecological combination ponds and implemented an interpretable machine learning approach to predict effluent total nitrogen concentration under variable influent quality pressure. Subsequently, the Shapley additive explanation method and partial dependence plots were introduced to analyze the contributions of different input variables to predictive outcomes, identifying key operational parameters that improve total nitrogen removal while balancing energy consumption and chemical oxygen demand dosage reduction. And then, a graphical user interface was established to allow operators to conduct prediction and parameters optimization through an intuitive interactive platform, simultaneous reductions in effluent total nitrogen concentration, energy consumption, and external carbon source usage were optimized. It is noteworthy that WWTPs are high energy-consuming facilities, accounting for 1 % to 2 % of total national carbon emissions in China, and they offer significant potential for carbon emission reduction (Sabia et al. 2020; Zhang et al. 2021; Dai et al. 2021). Therefore, this study calculated total carbon emission reductions solely based on enhanced total nitrogen removal, reduced energy consumption, and lower chemical oxygen demand dosage to evaluate the carbon reduction potential of wastewater treatment plant with ecological combination ponds.

2. Materials and methods

2.1. The ecological combination ponds operation overview

The full-scale urban wastewater treatment plant is located in Zhejiang, China. The wastewater treatment plant employing ecological combination ponds (ECPs) is designed with treatment capacity of 60,000 m^3/d. The ECPs is divided into 13 zones, and the average depth of each zone is 8.8 m (Fig. S1 and Fig. 1a). The first aerobic zone (1–5): 143,458 m^3, the anoxic zone (6–8): 97,075 m^3, the second aerobic zone (9–12): 142,266 m^3, and the settling zone (13): 74,800 m^3. One polymer external carbon source is added into the first anoxic zone (6) and the second anoxic zone (7), and mixed liquid is recycled from the settling zone back to the first anoxic zone (6). Influent quality and design effluent quality of full-scale urban wastewater treatment plant utilizing ECPs are illustrated in Table S1, and the effluent parameters (NH_4^+-N, COD, SS, pH, TP) are far below their national standards.

2.2. WWTP with ECPs and dataset information

In this study, daily operational data from a full-scale urban WWTP with ECPs were collected through extensive on-site monitoring and investigation. The dataset comprised 1095 sets of time-series data spanning from January 1, 2021 to December 31, 2023. The WWTP employing ECPs systematically monitored sixteen operationally critical parameters, including wastewater temperature, influent pH, and effluent chemical oxygen demand (COD) concentration. Through rigorous feature selection protocols, thirteen important parameters were identified as key process drivers for subsequent predictive modeling, including influent flowrate ($Flowrate_{inf}$), influent NH_4^+-N (NH_4^+-N_{inf}), influent total nitrogen (TN_{inf}), dissolved oxygen of Zone 5 (DO_5), total nitrogen of Zone 5 (TN_5), chemical oxygen demand of Zone 6 (COD_6), total nitrogen of Zone 6 (TN_6), dissolved oxygen of Zone 7 (DO_7), chemical oxygen demand of Zone 7 (COD_7), total nitrogen of Zone 7 (TN_7), external carbon source (ECS), energy consumption (EC), and effluent total nitrogen (TN_{eff}). These parameters were collected by the facility and the research team. Sampling locations included the center of each zone, as well as the inlet and outlet of wastewater in the WWTP. All parameters were measured with their respective methods described in Standard Methods (APHA, 2005).

Table S2 presents the dataset's performance metrics and characteristics, detailing the minimum, maximum, mean, standard deviation, and the 25 %, 50 %, and 75 % percentile values for each indicator. The results show a wide range of variability in the influent parameters (Table S1 and S2), with influent total nitrogen ranging from 8.6 to 108.0 mg/L (an average of 10.6 mg/L) and influent NH_4^+-N ranging from 5.4 to 62.7 mg/L (an average of 10.6 mg/L). Overall, wastewater quality indicators closely related to nitrogen removal exhibited significant fluctuations, highlighting the influent load challenges faced by the WWTP using ECPs while underscoring the expected complexities during model training. Moreover, the collected dataset was processed for missing values and outliers cleaning using KNN algorithm. As a preprocessing step, a small portion of records (<1 % of the entire dataset) containing outliers (data points significantly deviating from typical observations) was eliminated (Chen et al., 2024b).

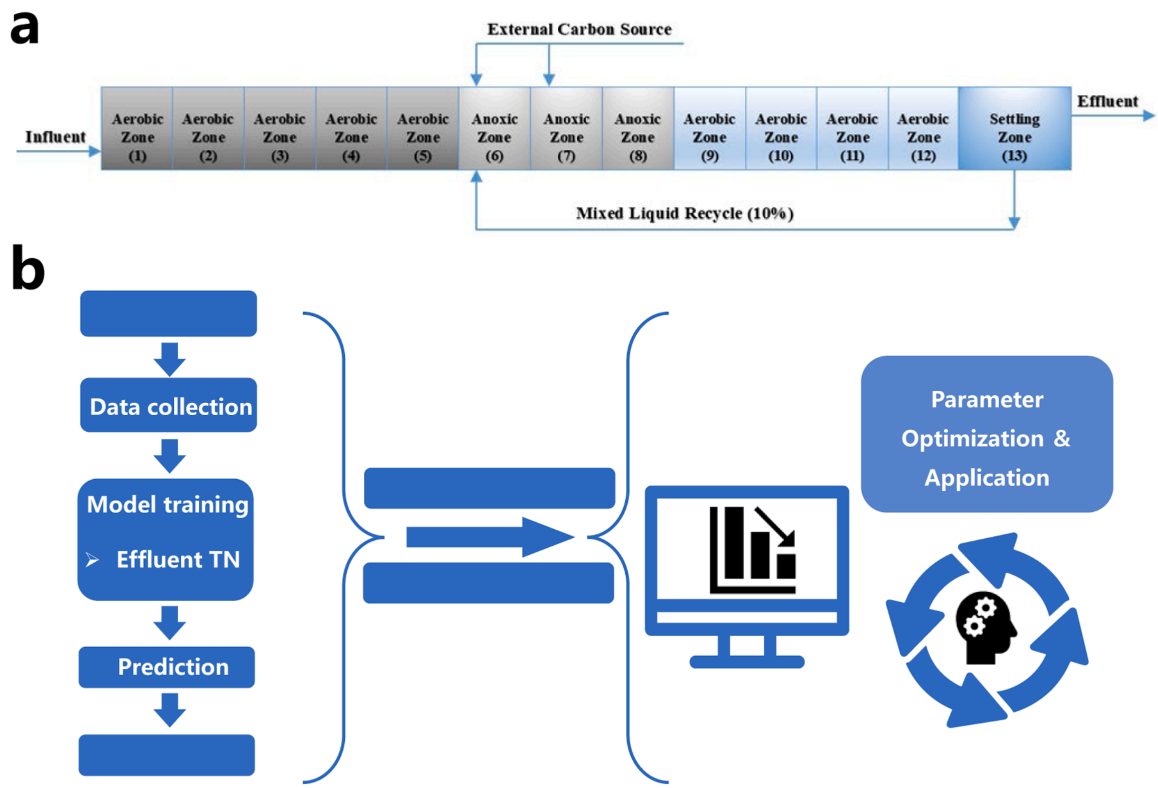

Fig. 1. (a) Simplified flowchart of ecological combination ponds in full-scale urban wastewater treatment plant; (b) ML framework for prediction and optimization of effluent TN concentration, and the optimal parameter combination was determined by SHAP analysis and PDP to enhance nitrogen removal while balancing energy consumption and COD dosage reduction.

2.3. Model development and evaluation

The preprocessed dataset was partitioned into an 80 % training set and a 20 % test set for model development. To enhance the model's generalization, Pearson correlation analysis was conducted to meticulously screen features with low predictive gains and collinearity. This strategy not only effectively reduced the feature count but also preserved the crucial information inherent in the original dataset. Concurrently, SHAP (SHapley Additive exPlanations) values were employed to systematically assess the significance of 16 initial features, including effluent TN, and less influentialfeatures were excluded to mitigate redundancy. These combined strategies enabled efficient dimensionality reduction. Ultimately, a refined dataset comprising 13 key features, including effluent TN, was constructed based on a balanced consideration of feature significance, correlation, and model complexity. Model development was conducted in the JupyterLab environment on the Anaconda platform using Python 3.8. To prevent overfitting and over-training, an Early Stopping strategy was applied. The training stopped when the validation set's mean squared error did not improve for 50 consecutive rounds, thereby preserving the model's generalization capability. Hyperparameter tuning was performed using the Grid Search method and 5-fold cross-validation was employed to thoroughly evaluate the model's performance and reduce the results' variance. For performance comparison and evaluation, two widely used metrics were utilized: determination coefficient (R^2, Eq. (1)) and root-mean-square error (RMSE, Eq. (2)). RMSE quantified the differences between the predicted and observed values, while R^2 assessed the model's goodness of fit, with higher R^2 values indicating more accurate predictions. Following hyperparameter optimization, the model was tested on a reserved dataset to compute the evaluation metrics, including R^2 and RMSE, to validate its performance. Based on model performance and interpretability, multiple machine learning algorithms were selected for comparison. Multiple linear regression (MLR), Support Vector Machines (SVM), Light Gradient Boosting Machine (LightGBM), Random Forest (RF), eXtreme Gradient Boosting (XGBoost), and categorical boosting (CatBoost) were used for training as prediction models. A detailed description of these six models is provided in Text S1. After optimizing the models of all six algorithms, the optimal models were compared to determine the best-performing model for testing TN removal. The confidence interval (CI) for the predicted values was computed using Eq. (3).

$$R^2 = 1 - \frac{\sum_{n=1}^{N} (\hat{y} - y)^2}{\sum_{n=1}^{N} (\hat{y} - \bar{y})^2} \tag{1}$$

$$RMSE = \sqrt{\frac{\sum_{n=1}^{N} (\hat{y} - y)^2}{N}} \tag{2}$$

$$CI = \bar{y} \pm z \frac{\sigma}{\sqrt{N}} \tag{3}$$

where y, $\bar{y}$, and $\hat{y}$ represent the actual target value, the mean target value, and the predicted target value, respectively. Additionally, z and σ represent the confidence level value and the standard deviation, while N denotes the total number of data points.

2.4. Optimization strategy for TN removal

Based on the prediction model's output results, targeted optimization strategies for nitrogen removal were developed by adjusting process parameters to enhance nitrogen removal efficiency. The SHAP method and partial dependence plots (PDP) were employed to assess the relative importance of the features used in the optimal models. SHAP is a method used to interpret machine learning model predictions by evaluating each feature's contribution to the model's output, thereby offering insights into the relative importance of individual features (Lundberg et al.

2018). Optimal Shapley values, derived from game theory, provide a comprehensive understanding of feature effects, shedding light on how these effects vary across individual observations (Song et al. 2020; Zhu et al. 2022). The key advantage of SHAP values is their capacity to quantify both the positive and negative contributions of features to model predictions while simultaneously considering the effects of individual variables and their interactions (Chen et al., 2024b). SHAP can be simplified by mapping to $x = h_x(z)$, allowing the original model $f(x)$ to be approximated as a linear function of the binary variables based on z, as illustrated in Eqs (4)-(6) below:

$$f(x) = g(z) + \varphi_0 + \sum_{i=1}^{M} \varphi_i z_i \tag{4}$$

where $z = \{0,1\}^M$, with M representing the number of input variables, and $\varphi_0 = f(h_x(0))$. The φ_i values denote feature contributions and are calculated as follows:

$$\varphi_i = \sum_{S \in F \setminus \{i\}} \frac{|S|!(M - |S|! - 1)!}{M!} [f_x(S \cup \{i\}) - f_x(S)] \tag{5}$$

$$f_x(S) = f\left(h_x^{-1}(z)\right) = E[f(x)|x_s] \tag{6}$$

where F denotes the set of non-zero input z, S represents the subset of F excluding the feature i^{th}, and φ_i is the unified measure of additive feature attributions, known as the SHAP value.

The mean absolute Shapley (MAS) value is calculated by averaging the absolute SHAP values of a feature across all data points, thereby quantifying the feature's importance to the model's prediction results (Lundberg et al. 2017). In this study, SHAP analysis was utilized to evaluate the importance of features in the regression tasks for predicting TN removal. The SHAP summary diagram further elucidates the sequence of factors influencing TN removal. PDP was employed to further understand the influence and correlation of each feature. Additionally, PDP permits the examination of the impact of interactive alterations between two features on the model prediction, resulting in a two-dimensional PDP (2D-PDP) (Deng et al. 2023; Gupta et al. 2023). For the analysis, the 'shap' and 'pdpbox' packages in Python were employed to calculate the SHAP and MAS values and to generate PDP for the features within the models. The optimal model corresponding to each dataset was designated as the explainer. SHAP analysis was conducted using the SHAP package, while PDP analysis is performed using the pdpbox package. The workflow diagram of this research is shown in Fig. 1b After using PDP to analyze the optimal parameter range for reducing TN removal and energy consumption simultaneously, ML framework was developed for the prediction and optimization of effluent TN concentration. The optimal parameters combination was determined through SHAP analysis to enhance nitrogen removal while balancing energy consumption and COD dosage reduction. To facilitate the practical application of the model, a user-friendly graphical user interface (GUI) was developed using the tkinter package (Palansooriya et al. 2022). In addition, a simplified Non-dominated Sorting Genetic Algorithm II (NSGA-II) was employed to address the multi-objective optimization problem among effluent TN, energy consumption, and external carbon source consumption (Chen et al., 2024b). The Technique for Order Preference by Similarity to an Ideal Solution (TOPSIS) method was utilized for analysis to obtain the optimal combination of operating parameters (Padrón-Páez et al. 2020). The detailed information is presented in Text S2.

3. Results and discussion

3.1. Model evaluation and comparison based on effluent TN concentration

Fig. S2 shows the training plots of six machine learning models, including MLR, CatBoost, SVM, LightGBM, RF, and XGBoost, with effluent TN concentration as the target feature (sixteen important parameters). While MLR demonstrated limited predictive capacity with test $R^2 = 0.459$ (training $R^2 = 0.497$), the remaining five nonlinear models achieved superior performance, yielding test R^2 values between 0.775 and 0.829. The XGBoost model exhibited optimal performance metrics, achieving near-perfect training accuracy ($R^2 = 0.999$, RMSE = 0.073) but showing weaker generalization on the test set ($R^2 = 0.829$, RMSE = 1.174). Consequently, the test R^2 value suggests potential limitations in generalizability, necessitating further optimization. Subsequently, SHAP analysis was conducted to quantify the feature contributions (Fig. S3). The results showed that the influent pH and influent NH_4^+-N exhibited the lowest feature importance value (0.020) and made a negligible contribution to improving the model's performance (Fig. S3a). Additionally, the influent pH remained stable within the range of 6.5 – 7.5, further supporting its classification as a redundant feature and subsequent exclusion. In contrast, the variable influent NH_4^+-N (5.4 – 62.7 mg/L) implied a potential association with effluent total nitrogen, thus justifying its retention.

The Pearson correlation coefficient was used to analyze the correlation between each feature and the effluent TN (Fig. S4). Both the effluent COD (0.24) and the wastewater temperature (-0.26) showed weak correlations with effluent TN, as well as low overall correlations with other features, with no obviously strong correlated features. Additionally, the effluent COD concentration (14 – 35 mg/L) remained well below the national discharge standard, indicating limited influence on TN_{eff} concentration. Moreover, in the actual operation of the wastewater treatment plants with ECPs, the adjustment of the wastewater temperature (17 – 30 °C) is limited. Therefore, wastewater temperature, influent pH, and effluent COD were excluded as redundant features to reduce the model complexity and enhance the generalization ability.

The optimized feature subset obtained through the pruning strategy was subsequently subjected to iterative retraining across multiple model architectures. Fig. 2 presents training plots of six machine learning models including MLR, SVM, RF, LightGBM, CatBoost, and XGBoost. The MLR model achieved low testing R^2 value of 0.603 and training R^2 value of 0.624. In contrast, the other five models attained high testing R^2 values, ranging from 0.829 to 0.911. These results indicate that a simple linear model is insufficient for predicting TN_{eff} concentration from multiple influent parameters and process parameters. The other five algorithms are then compared based on the R^2 and RMSE on the testing set and training set (Table S3). The XGBoost model recorded the highest performance, with R^2 values of 0.997 and 0.911, and RMSE values of 0.196 and 1.283 for training and testing sets, respectively. The results indicate that dimensionality reduction guided by the SHAP feature importance assessment significantly improved the model performance. The R^2 value of the testing set for the best – performing model (XGBoost) increased from 0.829 to 0.911.

Fig. 2f depicts the curve fitting of the XGBoost model, including a 95 % prediction band. Using the combined dataset of all 13 features, the XGBoost model achieved the best test performance with R^2 of 0.911 and RMSE of 1.283, surpassing the other models and establishing it as the optimal model for predicting TN removal. Notably, the majority of data points in the test set fall within the 95 % confidence interval, further demonstrating the reliability and robustness of the model. These results indicate that XGBoost effectively captures the influent water quality parameters, the nonlinear relationships between operating parameters, and accurately predicts TN_{eff} concentration. Therefore, the XGBoost model was adopted for subsequent calculations and analyses to predict TN_{eff} concentration.

3.2. Model sensitivity analysis with SHAP and PDP

The SHAP and PDP methods were employed to analyze feature importance, elucidate the prediction mechanisms of the XGBoost model, and identify key parameters influencing effluent TN concentration.

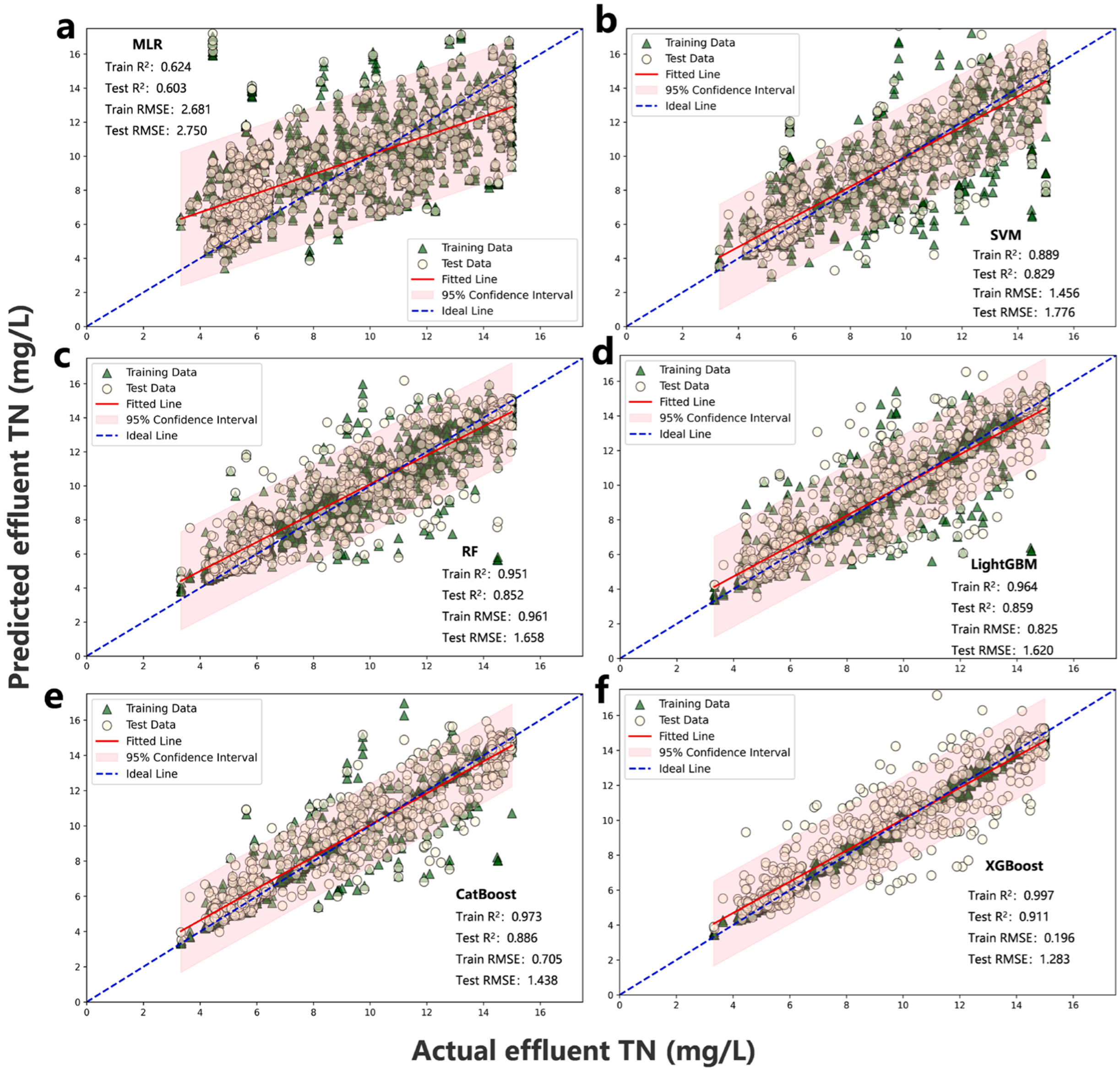

Fig. 2. Training plots of six machine learning models with effluent TN concentration as the target feature (a) MLR; (b) SVM; (c) RF; (d) LightGBM; (e) CatBoost; (f) XGBoost. Red lines denote the regression lines for the test sets, while pink areas represent the 95 % confidence intervals around these regression lines.

Figs. 3a and 3b present the MAS value ranking and the SHAP summary diagram for all features affecting TN_{eff}, respectively. In the SHAP summary graph, each point represents the SHAP value for a single data point within the model. The magnitude of the SHAP value, whether positive or negative, indicates the extent of its contribution to the predicted output. Subsequently, the average absolute SHAP value for each data point is calculated in the MAS graph, highlighting their importance to the model. The results revealed that TN_7, EC, DO_5, TN_6, $Flowrate_{inf}$, and DO_7 were the most influential features in determining TN_{eff} concentration. Although the ranking differs slightly between Fig. 3a and Fig. 3b, both identify TN_7, TN_6, EC, DO_5, and DO_7 as key parameters. Specifically, TN_7 and TN_6 are wastewater quality parameters, while EC, DO_5, and DO_7 are process parameters with controllable potential, categorizing them as key influential factors. Importantly, DO_5, and DO_7 are significantly related to EC, providing a logical foundation for our subsequent intelligent EC reduction strategies. These findings demonstrate that ML-based feature ranking offers a quantitative correlation between input feature and TN_{eff} output, thereby providing an effective strategy to understand the control mechanisms. In addition, since external carbon source is strongly correlated with COD_6 and COD_7, and potentially impacts on TN_{eff} concentration, it is essential to further investigate the relationship between COD dosage and TN_{eff} concentration, alongside focusing on the most significant contributors.

To further elucidate the effect of features on the model's output, PDP was used to investigate both the influence of single feature and the interaction between multiple features on effluent TN concentration. The TN_{eff} concentration can be influenced by both nitrification- and denitrification-mediated processes. Fig. 3c displayed the PDP for DO_5, and the results indicate that maintaining DO_5 concentrations achieves higher TN removal efficiency. Zone 5 functions as the aerobic zone, where lower DO concentration is unfavorable for microbial nitrification (Chen et al. 2018). It is noteworthy that aeration in Zone 5 also serves as a stirring role, preventing the DO_5 concentration from dropping too low, which represents a significant drawback of the ECPs process. The lack of DO_5 control further results in concentrations exceeding 10 mg/L. This challenge underscores the importance of employing machine learning methods to optimize DO_5 concentration and reduce power consumption.

Fig. 3d illustrates the PDP for DO_7, revealing a positive correlation with the TN_{eff} concentration. This is attributed to Zone 7 being an anoxic zone, where higher DO level impedes the denitrification efficiency of

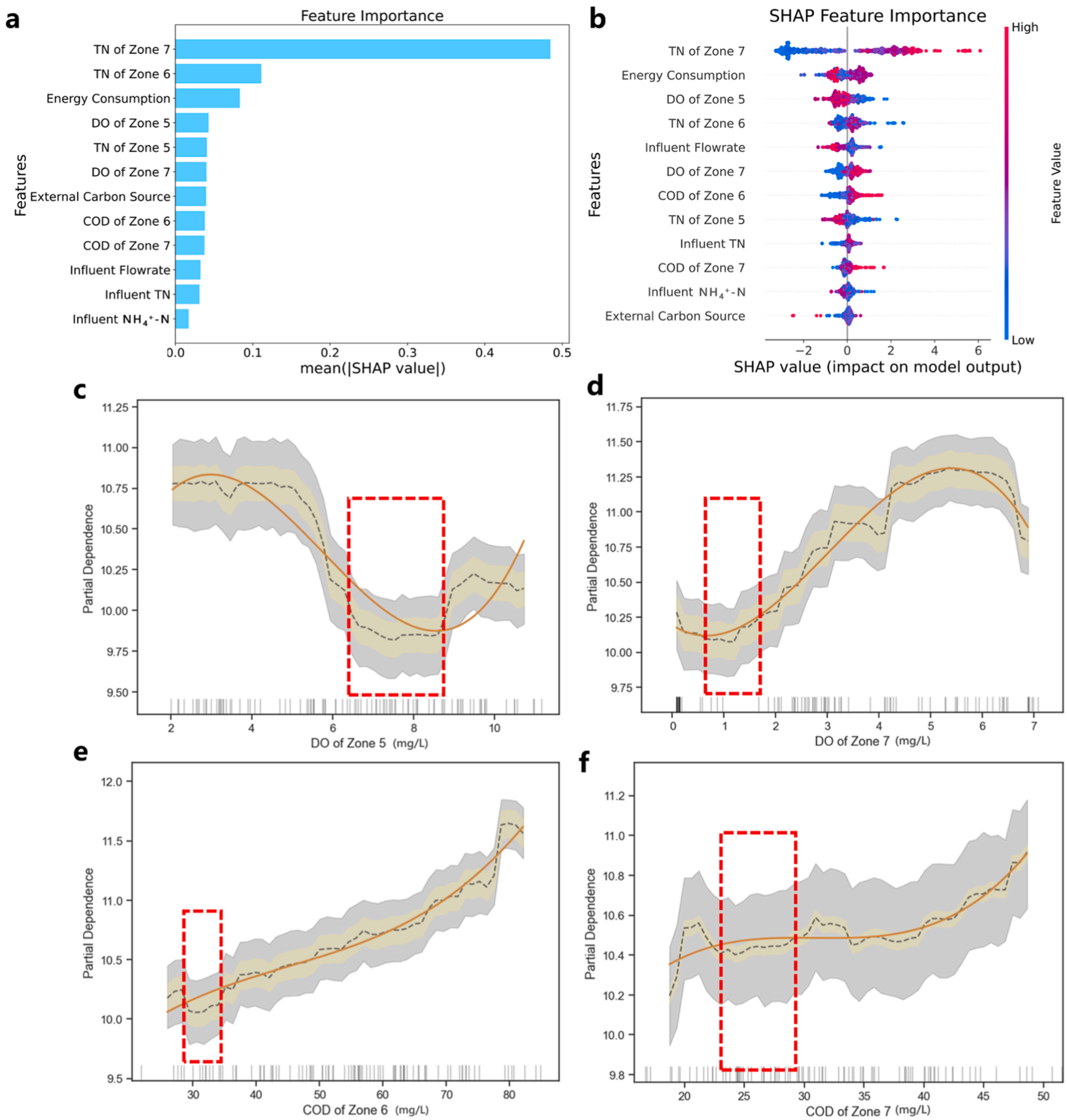

Fig. 3. (a) SHAP feature importance of the XGBoost model for predicting TN_{eff}; (b) SHAP significance analysis of the XGBoost model; (c) Partial dependence plots of the feature "DO of Zone 5″; (d) Partial dependence plots of the feature "DO of Zone 7″; (e) Partial dependence plots of the feature "COD of Zone 6″; (f) Partial dependence plots of the feature "COD of Zone 7". The dashed line represents the curve connected by the PDP values of each point. The solid line is a smoothed fit to the trend of the dashed line. The yellow shaded area is the 95 % confidence interval of the PDP values, and the gray area is the arrangement of the Individual Conditional Expectation (ICE) values for each sample. The short lines on the axis indicate the distribution of data on the corresponding feature of the X-axis. Red-dotted boxes highlight specific ranges of feature combinations linked to reduced effluent TN concentration.

microorganisms (Jin et al. 2023). Figs. 3e and 3f present the PDP for COD_6 and COD_7, which demonstrate that the appropriate COD dosages enhance TN removal performance. Conversely, excessive addition of COD increases both wastewater load and chemicals consumption, thereby negatively impacting the denitrification process (Pan et al. 2024). The PDP for additional carbon source further confirms the optimal carbon source dosing interval, revealing the potential for chemicals savings (Fig. S5). The PDP results for the other features illustrate the complexity of the effects of individual features on TN_{eff}, which further illustrates the necessity of using machine learning (Fig. S6).

Fig. 4a shows the interaction between DO_5 and DO_7 on TN_{eff} concentration. The analysis indicates that lower DO_7 combined with higher DO_5 levels achieve higher TN removal efficiency (He et al. 2018). Fig. 4b examines the interaction between COD_6 and COD_7 on TN_{eff} concentration. The findings reveal that appropriate COD concentration positively influence TN removal efficiency. Figs. 4c and 4d present the PDP diagrams for the interactions between ECS and COD_6, ECS and COD_7, respectively. These results indicate a conversion relationship between ECS and COD, suggesting that lower concentrations of ECS and COD contribute to reduced TN_{eff} concentration. To summarize, the optimal conditions for enhanced TN removal efficiency are as follows: DO_5 ranging from 5.8 to 8.8 mg/L, DO_7 ranging from 0.3 to 2.8 mg/L, COD_6 ranging from 28.0 to 35.5 mg/L, COD_7 ranging from 0 to 26.4 mg/L, and ECS dosage ranging from 1.9 to 2.3 m^3/d (Table S4).

3.3. Identifying and ranking the key parameters influencing TN removal performance

The ECPs in the WWTP were designed to achieve TN removal

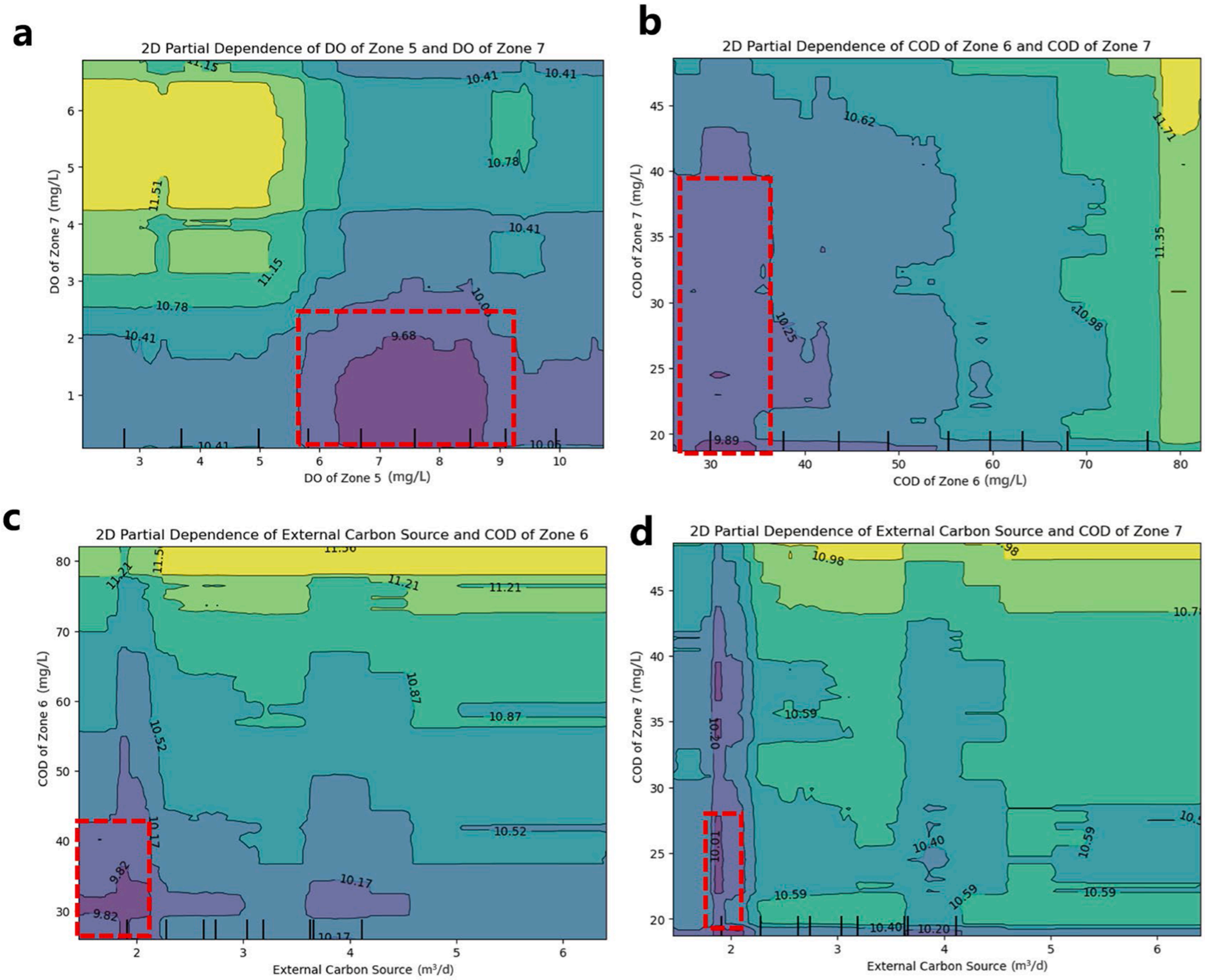

Fig. 4. (a) Partial dependence plots interact for features "DO of Zone 5″ and "DO of Zone 7″; (b) Partial dependence plots interact for features "COD of Zone 6″ and "COD of Zone 7″; (c) Partial dependence plots interact for features "External carbon source" and "COD of Zone 6″; (d) Partial dependence plots interact for features "External carbon source" and "COD of Zone 7". The numbers in the figure represent the values related to the effluent TN concentration under specific feature combinations. Darker-colored areas represent feature combinations that are more favorable for achieving lower effluent TN level, while red-dotted boxes highlight specific ranges of feature combinations linked to reduced effluent TN concentration.

through aerobic nitrification and anoxic denitrification. Accordingly, the XGBoost model was used for SHAP analysis to evaluate the significance of various features influencing TN removal performance. Fig. 3 illustrated the results from the analysis, TN_7, EC, and DO_5 were the three key parameters influencing TN removal efficiency. Zone 5 serves as the final aerobic zone where the DO concentration is uncontrolled, resulting in elevated DO level in the subsequent Zone 6 (the first anoxic zone). As a result, denitrification does not occur even when external carbon source is added into Zone 6. Similarly, in Zone 8 (the third anoxic zone), the absence of external carbon source prevents denitrification. In contrast, Zones 9–12 are aerobic zones that primarily convert residual ammonia nitrogen into nitrate nitrogen, contributing minimally to TN removal (Li et al. 2021; Daverey et al. 2015). However, in Zone 7 (the second anoxic zone), controlling DO concentration and adding external carbon source facilitate significant denitrification (Feng et al. 2018; Ma et al. 2017). Therefore, TN_7 is the most influential parameter affecting effluent TN concentration.

Furthermore, the uncontrolled DO concentrations in Zone 5 and Zone 7 result in excessive aeration, thereby increasing power consumption. Under fluctuating influent quality conditions, the unregulated dosing of external carbon source leads to excessive addition, raising chemicals consumption. The increased power consumption and chemicals usage elevate overall energy consumption. Accordingly, EC is the second most influential parameter on TN removal efficiency. Notably, the highest DO concentration exceeds 9.0 mg/L in Zone 5, and the lack of DO control leads to over-aeration. Although Zone 6 was designed as the first anoxic zone, it behaves aerobically due to the uncontrolled DO level in Zone 5. Consequently, DO_5 emerges as the third most significant parameter impacting TN removal performance.

3.4. *Relationship of nitrogen removal, energy consumption, and COD dosage*

Aeration and carbon sources significantly influence TN removal efficiency in ECP-based WWTPs by directly affecting the microbial nitrification and denitrification processes. Leveraging the insights from the 2D-PDP to uncover potential relationships among multiple features, Fig. 5a illustrates the combined effects of DO_5 and COD_6 on TN_{eff}. The predicted TN_{eff} reaches its minimum values when DO_5 is maintained between 6.3 and 8.8 mg/L, and COD_6 ranges from 29 to 35 mg/L. Additionally, the higher DO_5 concentration (above 9.5 mg/L) can also enhance TN removal but lead to excessive aeration, resulting in increased power consumption. The DO_7 concentration below 2 mg/L and COD_7 concentration around 19.5 mg/L positively influence TN removal efficiency (Fig. 5b). Similarly, TN_{eff} concentrations can be further decreased when COD_7 ranges from 34 to 39 mg/L. However, this

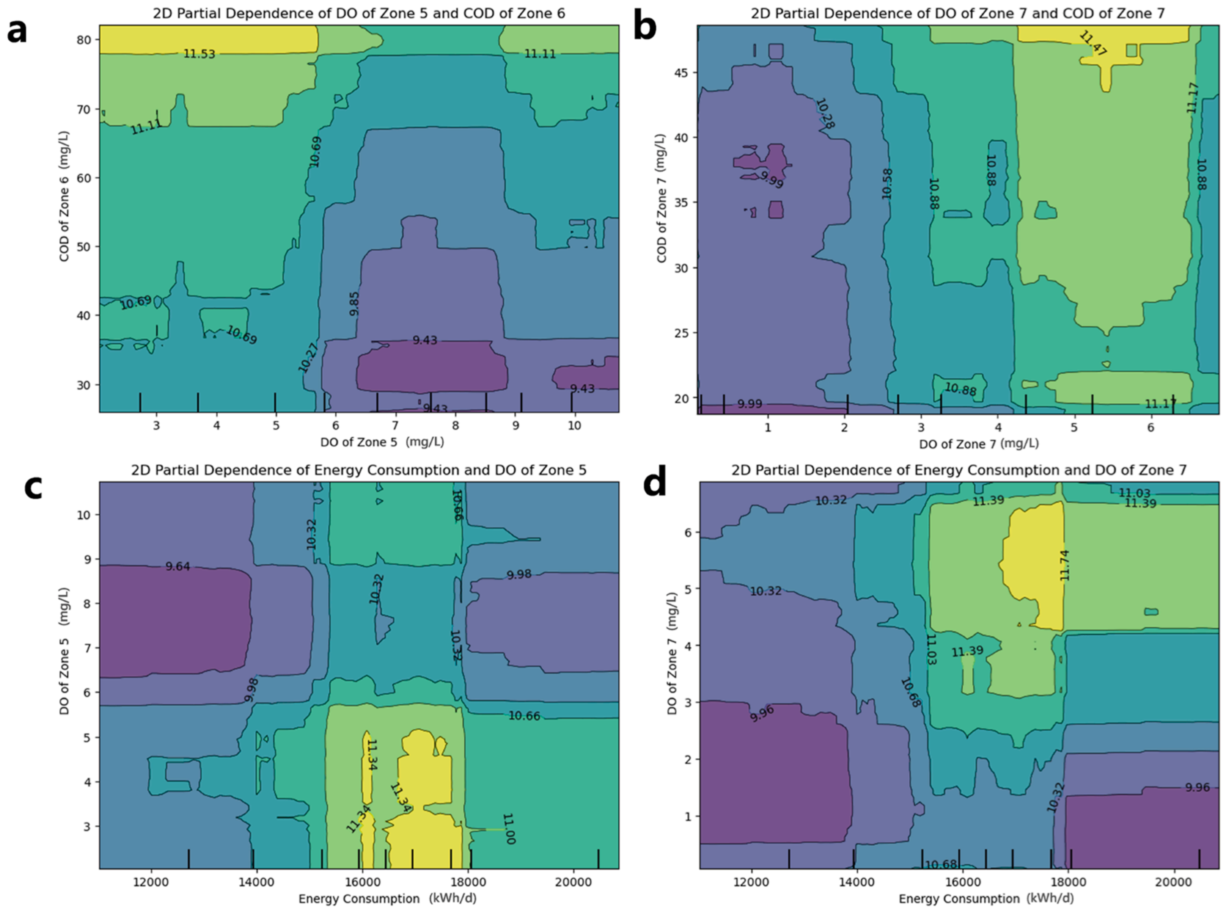

Fig. 5. (a) Partial dependence plots interact for features "DO of Zone 5″ and "COD of Zone 6″; (b) Partial dependence plots interact for features "DO of Zone 7″ and "COD of Zone 7″; (c) Partial dependence plots interact for features "Energy Consumption" and "DO of Zone 5″; (d) Partial dependence plots interact for features "Energy Consumption" and "DO of Zone 7". The numbers in the figure represent the values related to the effluent TN concentration under specific feature combinations. Darker-colored areas represent feature combinations that are more favorable for achieving lower effluent TN levels.

requires excessive ECS addition, leading to increased chemicals consumption.

Dissolved oxygen is regarded as the most critical factor influencing energy consumption in WWTPs utilizing ECPs. Fig. 5c demonstrates that TN_{eff} remains low when DO_5 ranges from 6.3 to 8.8 mg/L, with efficient TN removal achieved at lower energy consumption. Fig. 5d shows that TN_{eff} decreases when DO_7 is in the range of 0.5 – 3.0 mg/L, with effective TN removal occurring under both low and high energy consumption conditions. These findings underscore the substantial potential for energy savings in WWTP with ECPs. Furthermore, the optimized parameter ranges identified in Fig. 4 indicate that energy consumption is affected by the variations in both aeration and other parameters. Consequently, six ML models, i.e., SVM, MLR, RF, LightGBM, CatBoost, and XGBoost were developed to predict energy consumption (Fig. S7). The SVM model achieved low testing R^2 value of 0.523 and training R^2 value of 0.543. In contrast, the other five models attained higher testing R^2 values, ranging from 0.614 to 0.915. Among these, the XGBoost model demonstrated the best performance, achieving the highest R^2 of 0.915 and the lowest testing RMSE of 883.446. Therefore, the XGBoost prediction model is considered capable of capturing changes in energy consumption driven by variations in operating parameters.

3.5. Optimization of parameters for enhanced nitrogen removal while balancing energy consumption and COD dosage reduction

The ML model highlights the importance of optimizing process parameters to simultaneously lower TN_{eff} concentration, EC, and COD dosage. To facilitate practical application, a graphical user interface (GUI) integrating the TN_{eff}, EC and COD dosage model was developed (Fig. 6a). This GUI is based on the optimal control parameter interval derived from the PDP analysis and integrates the simplified NSGA-II and TOPSIS algorithms. By incorporating the optimal control strategy, users can adjust key operational parameters. Upon inputting influent wastewater quality data (e.g., influent NH_4^+-N, influent TN, TN concentrations in Zones 5, 6, and 7) and essential process parameters (e.g., influent flowrate, DO in Zones 5 and 7, COD in Zones 6 and 7, and ECS), the GUI can automatically predict and optimize effluent TN, energy consumption, and ECS usage.

Fig. S8 gives the practical application scenario analysis and optimization strategy of the GUI system. Data not involved in training and testing were collected to further validate the model through the GUI. In order to enhance TN removal while balancing energy consumption and COD dosage reductions, it is recommended to optimize the process parameters based on PDP interval. For example, the optimization of key parameters reduced the TN concentration in the effluent from 14.29 to 11.59 mg/L, energy consumption of WWTP decreased by 128.66 kWh/d, and COD dosage reduced by 34.29 mg/L, corresponding to a decrease in the external carbon source dosage from 3.72 to 2.25 m^3/d (1200 kg COD/m^3 ECS). Subsequently, the GUI automatically calculated enhanced TN removal, energy consumption, and COD dosage reductions based on influent data with varying characteristics over a 365-d period, achieving simultaneous reductions in effluent TN, energy consumption, and external carbon source usage. Notably, the effluent TN concentration decreased by 17.50 %, while COD dosage was reduced by 33.29 %

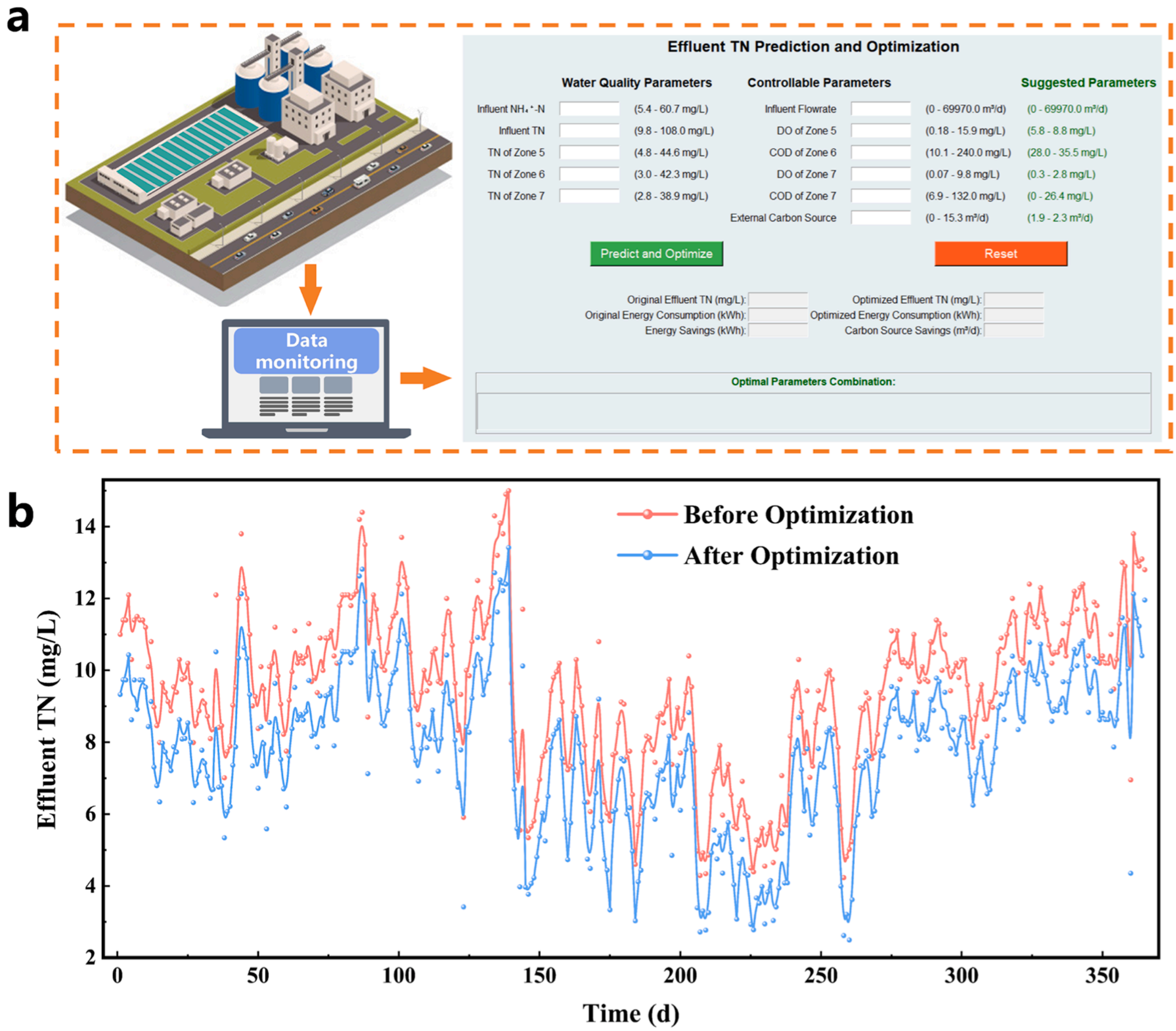

Fig. 6. Environmental application of machine learning model for enhanced nitrogen removal, energy savings, and COD dosage reduction. (a) Application guidance and parameters optimization of the machine learning model; (b) The effluent TN concentration before and after the optimization of operation parameters within one year.

annually. Fig. 6b shows a comparison of the effluent TN concentrations before and after optimization over the 365-day period. The TN values after optimization were consistently lower than those before optimization, indicating that the parameter combinations provided by the GUI effectively improved TN removal.

Energy consumption is a crucial factor in nitrogen removal optimization in WWTPs (Rosso et al. 2005). Fernández-Arévalo et al. have established predictive models for effluent TN and energy consumption, simultaneous optimization of these two factors was not been attained (Fernández-Arévalo et al. 2017). The Non-dominated Sorting Genetic Algorithm II model was employed for the multi-objective optimization of effluent chemical oxygen demand and total energy consumption, resulting in reductions of respectively 15 % and 18 %, achieving synergy between water quality improvement and energy saving (Chen et al., 2024b). In this study, enhanced TN removal by adjusting process operational parameters can effectively balance energy consumption and COD dosage reduction, achieving coordinated optimization. These results indicate that the optimal control strategy effectively improves TN removal in WWTP employing ECPs, while simultaneously reducing energy consumption and external carbon source usage.

3.6. *Carbon emission reduction in the WWTP with ECPs*

Carbon emission is an essential indicator for evaluating environmental impacts. Abulimiti et al. have explored the effect of energy-saving measure (e.g., DO control) on greenhouse gas emission (Abulimiti et al. 2022). Recent studies have applied the multi-objective optimization control to reduce greenhouse gas emission without increasing operating costs while maintaining acceptable effluent quality (Chen et al., 2024b; Sweetapple et al. 2014). This study achieved enhanced TN removal in a full-scale urban WWTP with ECPs. The key parameters were identified to improve TN removal while balancing energy consumption and COD dosage reduction. Consequently, the WWTP with ECPs demonstrated substantial potential for carbon emission reduction. To evaluate the potential, both direct and indirect carbon emissions were estimated. The calculation boundary was confined to the core treatment process, from influent entry to effluent discharge into the receiving water body. In accordance with IPCC guidelines, CO_2 emissions generated by microbial activity during biological treatment were excluded (IPCC. 2006; IPCC. 2019). The direct emission primarily includes nitrous oxide (N_2O) emission generated during nitrogen removal, and indirect emission mainly includes carbon dioxide (CO_2) emission of energy consumption and COD dosage. Total annual carbon emission reductions were calculated solely based on enhanced TN removal, along with reductions in energy consumption and COD dosage. The standard emission factors adopted in this study were as follows: 0.005 kg N_2O/kg N for nitrogen removal, 0.7035 kg CO_2/kWh for energy consumption, and 1.54 kg CO_2/kg COD for external carbon source usage. The detailed calculation process of carbon emission reduction is provided in the Supplementary Information (Text S3 and Table S5).

As illustrated in Table 1, total annual carbon emission reduction in urban WWTP utilizing ECPs is 788.40 t CO_2/y. Direct carbon emission reduction is 59.23 t CO_2/y, representing 7.51 % of the total decrease. Indirect carbon emission reduction includes 13.99 t CO_2/y from decreased energy consumption and 715.18 t CO_2/y from reduced COD dosage, accounting for 92.48 % of the overall reduction. Therefore, the optimization of enhanced total nitrogen removal while balancing energy consumption and COD dosage reduction in WWTP using ECPs by ML model is significant for achieving energy savings and carbon emission reduction goals. Notably, the WWTP employing ECPs in this study does not include the anaerobic zone as well as sludge treatment and disposal infrastructure, therefore, CH_4-related carbon emissions were not considered. Additionally, since the treatment process does not involve fuel and heat consumption, corresponding fuel-related and heat-related carbon emissions were also excluded. Accordingly, while the findings provide valuable insights into the carbon emission reduction potential of WWTP with ECPs, the limitations of the carbon emission estimation methods used should be considered when applying the results to other WWTPs or regions.

4. Conclusions

In this study, applying an interpretable machine learning approach to predict and optimize enhanced TN removal in urban WWTP utilizing ECPs under unstable influent quality pressure proved to be an effective method, substantially increasing the sustainability and effectiveness of wastewater treatment in small towns. A user-friendly GUI was developed to facilitate ongoing prediction and coordinated optimization of operational parameters (Fig. S8), allowing users to input process parameters and automatically predict as well as optimize TN removal, energy consumption, and COD dosage (https://github.com/Tomriddle-lab/TNeff-GUI/blob/master). By integrating the proposed optimization control strategies based on PDP analysis, users can adjust key process parameters to achieve the improvement in TN removal, energy savings, and COD dosage reduction. It is noteworthy that the ML-based optimization of enhanced nitrogen removal in a full-scale urban WWTP with ECPs represents a significant step forward. This approach contributes to the effective application of artificial intelligence in advancing carbon emission reduction of WWTPs and sustainable development goals with environmental benefits. Moreover, the collection, processing, and data mining of large datasets hold great promise for further advancements.

However, the current data collection process of WWTPs employing ECPs is insufficient. Based on the findings of this study, further optimization is recommended. For instance, sampling locations should include the center, inlet, and outlet of each zone, with averaged values used for analysis. Additionally, instead of using portable DO and pH meters, online DO probes and pH sensors should be installed at the center of each zone to enable real-time monitoring. Aeration intensity should be regulated based on DO concentrations in the respective aerobic and anoxic zones. These insights could help other existing or newly constructed WWTPs with ECPs improve their data collection and process control systems. While the XGBoost model developed in this study demonstrated strong performance for the full-scale urban WWTP utilizing ECPs, its generalization to other WWTPs with ECPs requires further investigation. The variations in WWTP configurations can significantly influence the accuracy and reliability of the model, and the effectiveness of ML approaches is inherently constrained by data quality. To address these limitations, additional model optimization strategies could be employed to enhance adaptability to configuration changes. For instance, the integration of association rule mining and principal component analysis may help elucidate complex relationships among influent wastewater quality characteristics, process variables, and effluent TN, thereby facilitating the identification of key influencing factors. Furthermore, incorporating comprehensive microbial community diversity data may further improve the model's transferability and applicability to other WWTPs.

Table 1
Carbon emission reduction in urban WWTP with ECPs.

Items	Specific value (t CO_2/y)
Direct carbon emission reduction	59.23
Indirect carbon emission reduction (energy consumption)	13.99
Indirect carbon emission reduction (COD dosage)	715.18
Total carbon emission reduction	788.40

CRediT authorship contribution statement

Jinhu Yun: Data curation, Formal analysis, Visualization, Software, Methodology, Conceptualization, Writing – original draft, Writing – review & editing. **Yang Yu:** Resources, Data curation. **Chenliang Tao:** Visualization, Software, Methodology. **Mudi Zhai:** Data curation, Formal analysis. **Hongliang Zhang:** Supervision. **Ying Chen:** Supervision. **Hongjing Li:** Resources, Project administration, Funding acquisition, Conceptualization, Writing – original draft, Writing – review & editing, Supervision. **Bao Zhang:** Resources. **Jun Ma:** Supervision.

Declaration of competing interest

The authors declare no competing financial interest.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2025.123976.

Data availability

All the code and the raw data used to complete this work are available at https://github.com/Tomriddle-lab/TNeff-GUI/blob/master.

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